LESSON 2 • TERM 4 • LEVEL 1

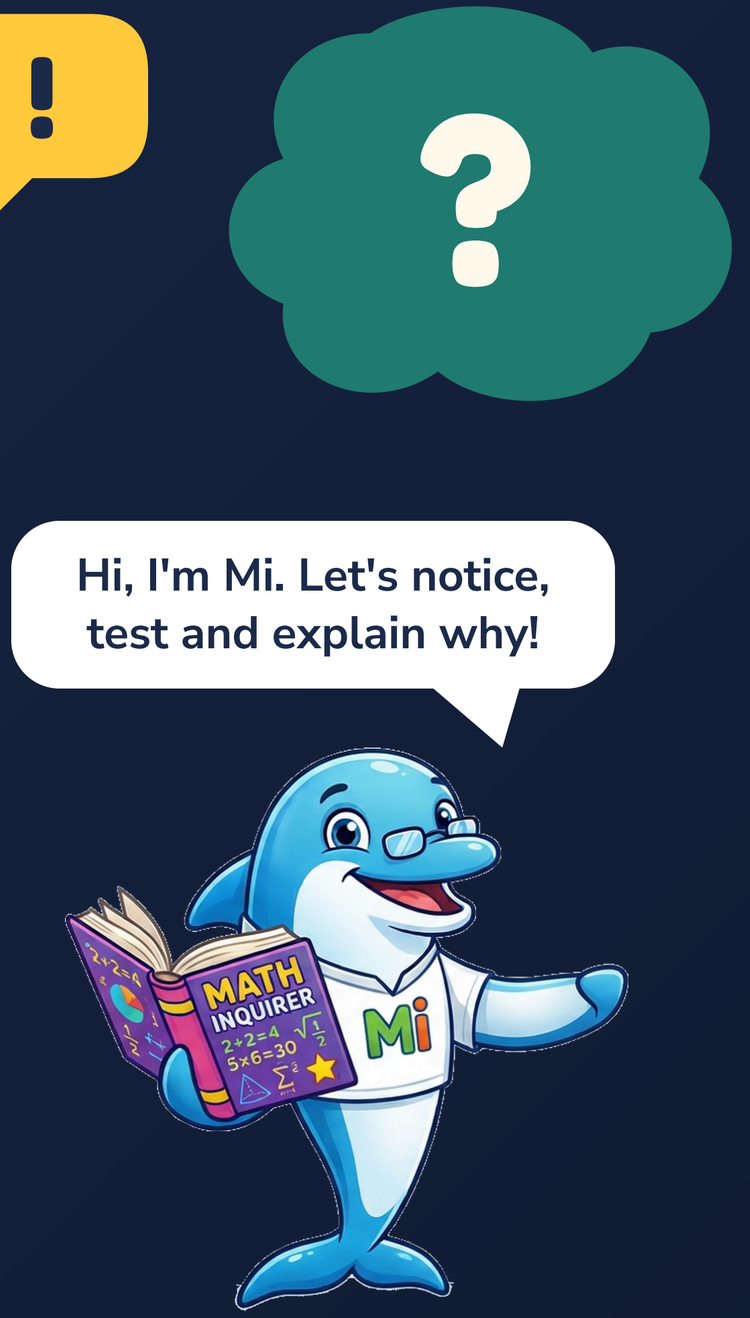
Build the Figure

Big shapes from small shapes

TODAY WE WILL

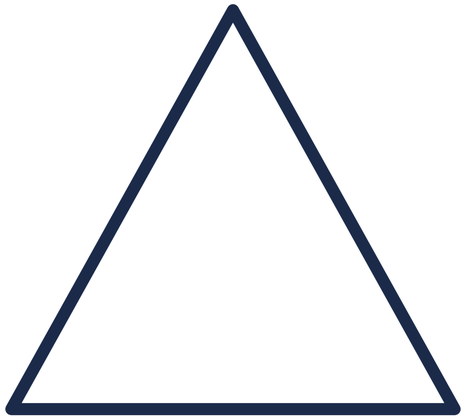
Put shapes together.

Find the shapes inside a figure.

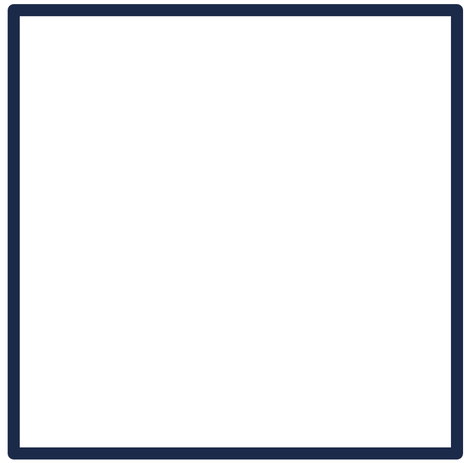


Which shape am I?

2–8 MIN



A



B



C

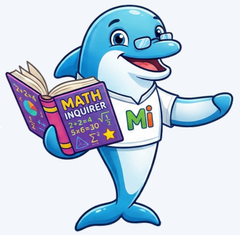
Which shape has 3 corners? Which has 4 sides that are all the same? Which has 2 long sides and 2 short sides?

Fold a square corner to corner

2–8 MIN

Take a paper square. Fold it corner to corner.

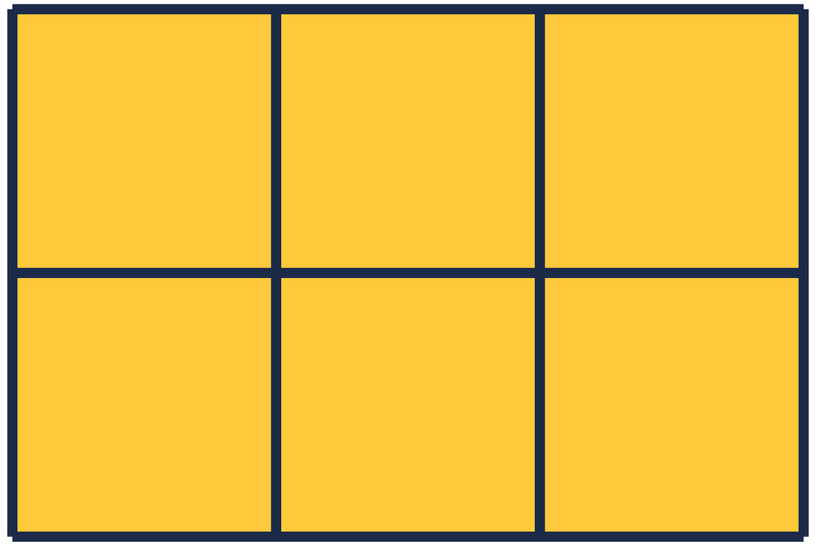
Open it. What 2 shapes do you see?



Fold it carefully!

Count by rows

2–8 MIN



How many squares make the rectangle? Count by rows.

Sneak-peek puzzles

8–12 MIN

PUZZLE 1: anyone can try

Look at the picture. How many triangles are there?



PUZZLE 2: think

Put 2 triangles from your kit together so that whole sides touch. Name one shape you can make. Draw it.

Explore: 2 triangles

12–24 MIN

1. Take 2 triangles from your kit.
2. Put them together so whole sides touch.
3. Make a square.
4. Now make a bigger triangle.

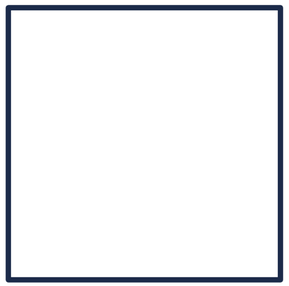
Whole sides must touch.

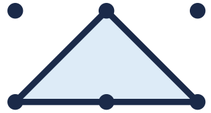
Explore: 2 triangles

12–24 MIN

1. Take 2 triangles from your kit.
2. Put them together so whole sides touch.
3. Make a square.
4. Now make a bigger triangle.

Whole sides must touch.





2 TRIANGLES MAKE

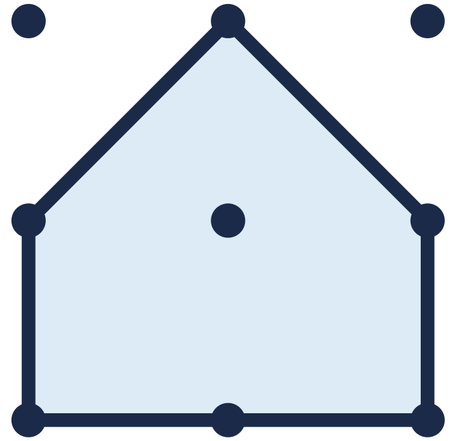
a square.

a bigger triangle.

Explore: build the house

12–24 MIN

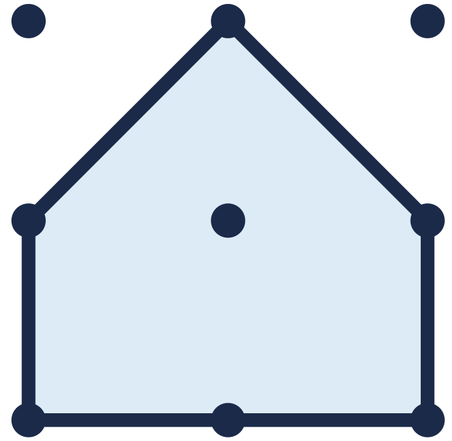
1. Look at the house.
2. Build it with your squares and triangles.
3. Which pieces did you use?



Explore: build the house

12–24 MIN

1. Look at the house.
2. Build it with your squares and triangles.
3. Which pieces did you use?



THE HOUSE USES

2 squares and 2 triangles.

That is 4 pieces.

Explore: your own picture

12–24 MIN

1. Make a picture with your pieces: a robot, a rocket or a boat.
2. Use at least 3 different kinds of shape.
3. Count how many of each shape you used.
4. Say it: My picture has … squares and … triangles.

Keep it on the table. Say it. Nobody holds it up.

2 shapes make 1 shape

24–36 MIN

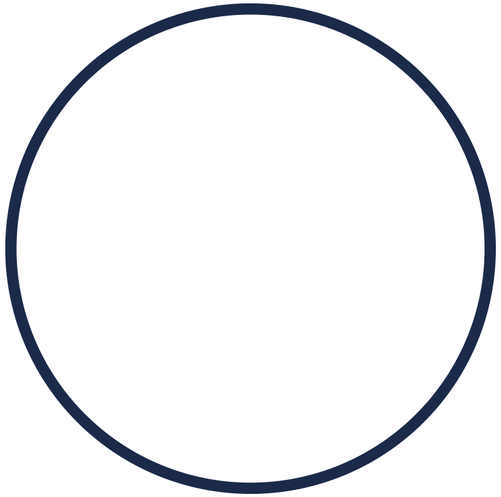
2 SQUARES

make a rectangle.



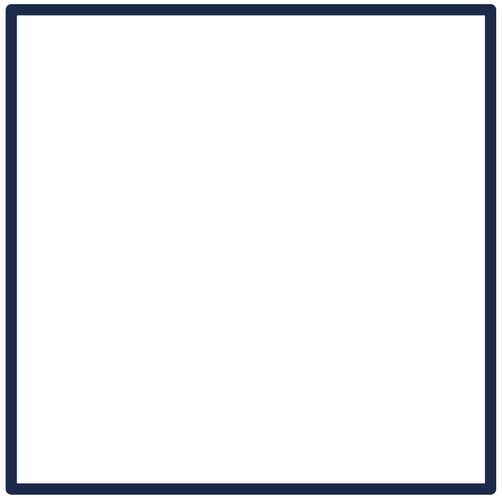
2 HALF CIRCLES

make a circle.



2 TRIANGLES

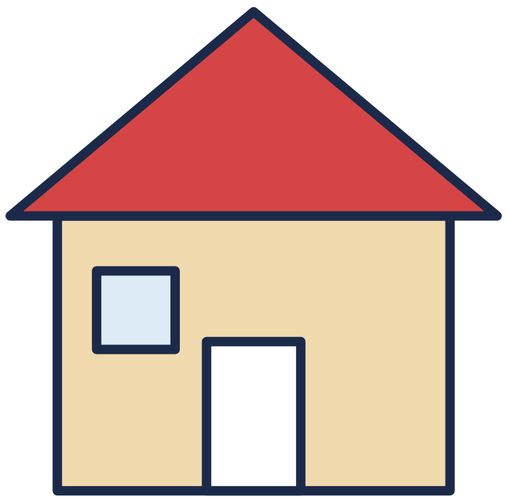
make a square or a bigger triangle.



Find the shapes inside

24–36 MIN

Look at the house. Count each kind of shape.



How many triangles?

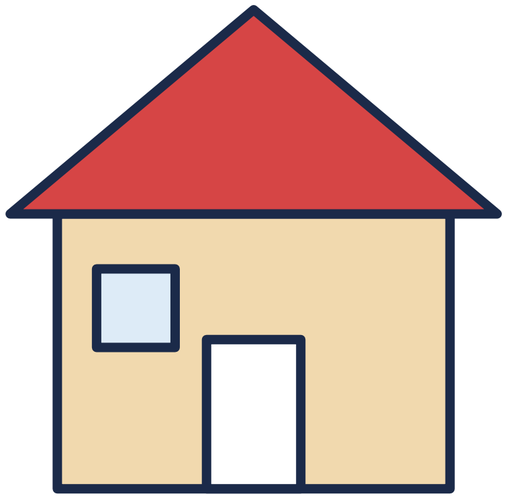
How many rectangles?

How many squares?

Find the shapes inside

24–36 MIN

Look at the house. Count each kind of shape.



How many triangles?

How many rectangles?

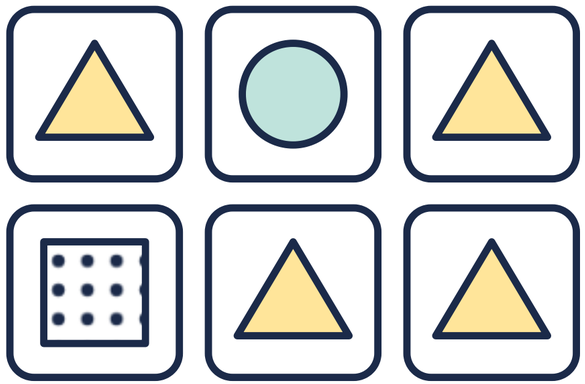
How many squares?

1 triangle, 2 rectangles and 1 square. The roof is a triangle. The wall and the door are rectangles. The window is a square.

Count each kind

24–36 MIN

Count each kind. Which kind is there most of?

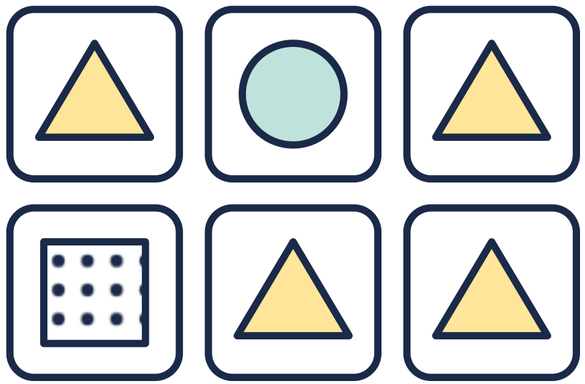


How many triangles? How many circles? How many squares?

Count each kind

24–36 MIN

Count each kind. Which kind is there most of?



How many triangles? How many circles? How many squares?

4 triangles, 1 circle and 1 square. There are 6 shapes in all. There are most triangles.

A rectangle can replace 2 squares

24–36 MIN

4 SQUARES

make a long rectangle.

That is 4 pieces.



2 RECTANGLES

make the same long rectangle.

That is 2 pieces.

A rectangle is as long as 2 squares.

Your turn

24–36 MIN

ITEM 1

A picture has 3 circles and 2 squares. How many shapes are there?

ITEM 2

Can 2 squares make a triangle? Say Yes or No.

Your turn

24–36 MIN

ITEM 1

A picture has 3 circles and 2 squares. How many shapes are there?

ITEM 2

Can 2 squares make a triangle? Say Yes or No.

Item 1: 5 shapes. Item 2: No. 2 squares make a rectangle, not a triangle.

36–40 MIN

Shape moves

Stand up. Make each shape I call.

Triangle. Rectangle. Circle.

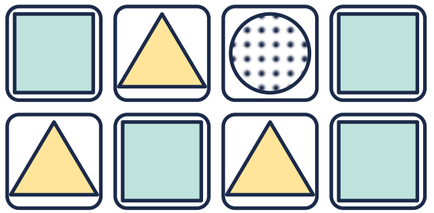
Sip some water.

Workbook page 1: Start here

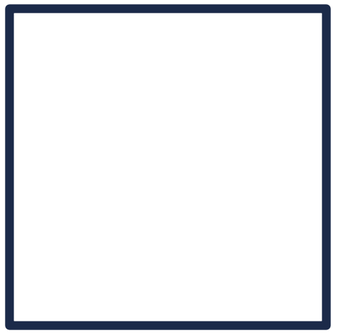
40–70 MIN

Questions 1 and 2. Take your time.

S1 Look at the picture of 8 shapes. Count each kind. Fill in the blanks.



S2 A square is cut into 2 triangles. Use your ruler. Draw the line.



Check page 1

40–70 MIN

S1	4 squares; 3 triangles; 1 circle; 1 more square than triangles
S2	a line from one corner to the opposite corner; half

Workbook page 2: Start here

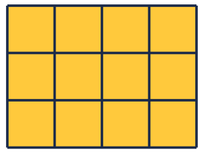
40–70 MIN

Questions 3 to 5. Take your time.

S3 Build the boat with squares and triangles. Use as few triangles as you can. Fill in the blanks.



S4 The rectangle is made of squares. Fill in the blanks.



S5 Fill in the blanks. Circle one word each time.

Check page 2

40–70 MIN

S3	2 triangles; 2 squares
S4	4; 3; 12
S5	rectangle; circle; half circle

40–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Split a figure. Every Triangle
Go further (page 4)	The fewest pieces
Investigation (page 5)	2 triangles, many shapes
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

A CHILD SAYS

2 triangles always make a square.

Is the child right? Test it with your triangles.

I think ... because ...

Back to the sneak-peek

70–76 MIN

Puzzle 1: 2 triangles. Puzzle 2: a square, a bigger triangle or a slanted four-sided shape. Many answers are right.

Exit ticket

76–82 MIN

On your own. No looking back.

E1 **Mei makes a picture with 3 squares, 2 triangles and 2 circles. How many shapes are in the picture? Fill in the blank.**

There are ☐ shapes in the picture.

E2 **Can 2 triangles make a square? Circle Yes or No.**

E3 **A rectangle is made of 6 squares. Mei makes it again with rectangles that are each 2 squares long. How many rectangles does she need?**

She needs ☐ rectangles.

Check the exit ticket

76–82 MIN

E1	7
E2	Yes
E3	3

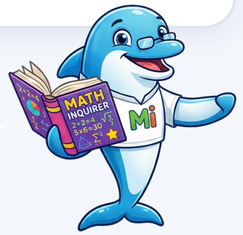
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ putting pieces together ☐ counting each kind ☐ splitting a figure ☐ something else
What helped me	☐ my shape pieces ☐ touching each shape as I count ☐ checking my answer ☐ I was not stuck
Next time I will try	☐ trying the pieces first ☐ counting one kind at a time ☐ checking my answer ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

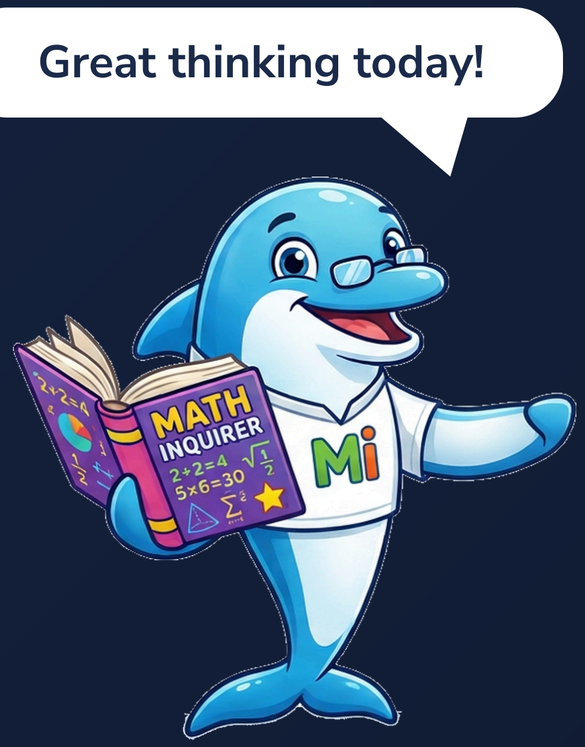
Well done, inquirers!

Show a grown-up one figure you built today.

NEXT LESSON

Copy on the Grid

Have ready: the grid page (it is not cut), a button for a counter, a ruler and a pencil.



OPTIONAL

Optional: 4 squares, one rectangle

Mei has 4 squares. She makes a square with all 4.

What shape is it? Can she make it with fewer pieces?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Do 2 squares always make a rectangle?
Make a puzzle	Build a figure. Ask: how many of each shape?
Teach it back	Show a toy how 2 triangles make a square.

Optional. Everyone may join.